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10 CFR 50.73

GNRO2022/00004

January 19, 2022

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Grand Gulf Nuclear Station, Unit 1 Licensee Event Report 2021-004-00,
Procedure Inadequacy Resulted in Core Monitoring System
Miscalibration and Violation of Technical Specification

Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
Renewed License No. NPF-29

Attached is Licensee Event Report 2021-004-00, Procedure Inadequacy Resulted in Core Monitoring System Miscalibration and Violation of Technical Specification. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B) any operation or condition which was prohibited by the plant's Technical Specifications.

This letter contains no new Regulatory Commitments. Should you have any questions concerning the content of this letter, please contact Jeff Hardy, Regulatory Assurance Manager at 802-380-5124.

Sincerely,

A handwritten signature in blue ink, appearing to read "JAH", is positioned to the left of the printed name.

Jeff
Hardy

Digitally signed
by Jeff Hardy
Date: 2022.01.19
15:21:23 -06'00'

JAH/saw

Attachments: Licensee Event Report 2021-004-00

cc: NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

U.S Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Attachment
Licensee Event Report 2021-004-00



LICENSEE EVENT REPORT (LER)

(See Page 3 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<https://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collection Branch (T-6 A10M), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; e-mail: cira_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Grand Gulf Nuclear Station, Unit 1	2. Docket Number 05000 - 416	3. Page 1 of 3
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4. Title
Procedure inadequacy resulted in Core Monitoring System miscalibration and violation of technical specification

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
11	22	2021	2021	004	00	01	19	2022	N/A	N/A
									N/A	N/A

9. Operating Mode 1	10. Power Level 100
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☒ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)	
☐ 20.2203(a)(2)(iii)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.77(a)(2)(ii)
☐ 20.2203(a)(2)(iv)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	
☐ 20.2203(a)(2)(v)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	

☐ Other (Specify here, in Abstract, or in NRC 366A).

12. Licensee Contact for this LER

Licensee Contact Jeff Hardy, Manager Regulatory Assurance	Phone Number (Include Area Code) 802-380-5124
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable To IRIS	Cause	System	Component	Manufacturer	Reportable To IRIS
D	N/A	N/A	N/A	Y	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year
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16. Abstract (Limit to 1560 spaces, i.e., approximately 15 single-spaced typewritten lines)

On November 18, 2021, a Past Operability Evaluation determined that an issue associated with jet pump summer miscalibration was a reportable event. Specifically, while performing, 17-S-02-5, Post Refueling Recirculation System Flow Instrumentation Calibration (site procedure), it was discovered that multiple conversion calculations in the procedure were incorrect and the procedural guidance for which data points to use in the calculation was unclear. This affects the transmitter and computer point calibration data used by Instrumentation and Control (I&C) personnel. This caused an incorrect core flow indication to the control room panel indication and the Core Monitoring System. Resulting in a violation of Technical Specification (TS) 3.2.2 for Minimum Critical Power Ratio (MCPR).

There were no actual safety consequences for this event. Reactor Engineering and Operations had agreed to a Compensatory Measure and issued Standing Order 21-05 (Administrative Limit for Maximum Fraction of Limiting Core Power Ratio MFLCPR) to establish an administrative reduction in MFLCPR limit to 0.970. Standing Order 21-018 (Total Core Flow Indication) was issued to provide operators the ability to calculate total core flow using reactor recirculation drive flow indications that were not impacted by the calibration of the jet pump instrumentation.

There were no consequences to the general safety of the public, nuclear safety, industrial safety, or radiological safety. No radiological releases occurred due to this event.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Grand Gulf Nuclear Station, Unit 1	05000 - 416	YEAR	SEQUENTIAL NUMBER	REV NO.
		2021	- 004	- 00

NARRATIVE**Plant Conditions:**

Grand Gulf Nuclear Station (GGNS) Unit 1 was operating at 100 percent power in MODE 1. There were no Structures, Systems, or Components that were inoperable that contributed to this event.

Event Description:

On January 29, 2021, due to errors in the site procedure, Reactor Engineering provided I&C personnel incorrect voltage inputs that resulted in the miscalibration of the Jet Pump Loop Flow Summers for Loop A and Loop B. This miscalibration caused an incorrect core flow indication to the control room panel indication and the Core Monitoring System, resulting in a violation of Technical Specification (TS) 3.2.2 MCPR.

On July 29, 2021, it was determined that multiple conversion calculations in the procedure were incorrect, and the procedural guidance was unclear. This resulted in the wrong data for jet pump flow being used to calibrate Jet Pump Loop Flow Summers for Loop A and Loop B. Additionally, it was determined that the core flow signal from the jet pump summers to the core monitoring software was 1.95 [Mlbm/hr] higher than actual core flow (a difference of approximately two percent of rated flow) and there was an adverse effect on thermal limits generated by the core monitoring software to satisfy technical specification Limiting Condition for Operation 3.2.2. In response to this issue, an administrative reduction in MFLCPR limit to 0.970 was applied. This was put in place via Standing Order 21-015.

On November 18, 2021, a Past Operability Evaluation determined that an issue associated with jet pump summer miscalibration was a reportable event. Specifically, while performing the site procedure, it was discovered that multiple conversion calculations in the procedure were incorrect and the procedural guidance for which data points to use in the calculation was unclear. This affects the transmitter and computer point calibration data used by I&C personnel. This caused an incorrect core flow indication to the control room panel indication and the Core Monitoring System Resulting in a violation of Technical Specification (TS) 3.2.2 for MCPR.

The Human Performance Evaluation identified two factors that contributed to the condition. First, the procedure used for performing the core flow instrumentation calibration, contains gaps in process as well as confusing and vague steps for calculations. The other factor identified was gaps in knowledge of the core flow calibration process, including gaps in knowledge of the methodology used in the jet pump calibration spreadsheet used with the site procedure to perform the calculations.

The Organizational and Programmatic (O&P) process identified two factors that contributed to the condition – process weaknesses and organizational problems with program execution. Procedure contains errors and there are inconsistencies between it and the spreadsheet used to perform the calculations. The procedure also contains gaps in the core flow calibration process compared to the standard method from the General Electric Station Nuclear Engineering Manual, and it is unclear and vague on certain steps performing calculations. There are also inconsistencies between the procedure and the spreadsheet used to perform the calculations. The responsibility to implement the Software Quality Assurance (SQA) program is not well understood for utilizing spreadsheets in the reactor engineering group.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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Grand Gulf Nuclear Station, Unit 1	05000 - 416	YEAR	SEQUENTIAL NUMBER	REV NO.
		2021	- 004	- 00

Safety Assessment:

There were no actual consequences for this event. Reactor Engineering and Operations agreed to a Compensatory Measure and issued Standing Order No: 21-015 to establish an administrative reduction in MFLCPR limit to 0.970. Standing Order 21-018 was issued to provide operators the ability to calculation total core flow using reactor recirculation drive flow indications that were not impacted by the calibration of the jet pump instrumentation. As a result, there was no impact to the health and safety of the public or plant personnel from this condition. In addition, this event does not meet the criteria for a Safety System Functional Failure.

Event Cause(s):

The site procedure, and the spreadsheet used to implement the procedure contain errors and gaps in the core flow calibration process.

The spreadsheet used by Reactor Engineering to perform the Core Flow Calibration process is not in the SQA Program.

There is a knowledge gap in Reactor Engineering of the Core Flow Calibration process.

The Apparent Cause Analysis from CR-GGN-2018-8553 had inadequate corrective actions. The 2018 evaluation determined that a lack of questioning attitude was a causal factor in that it was not applied to question the validity of the results. Corrective actions completed included revising the site procedure to compare new data to past data. However, the evaluation and corrective actions did not go a step further to question the procedure and process used. In the revision (12), the computer code program (JRPUMP) was retired, and the procedure was revised to perform the calibration without it. The 2018 revision did not clearly and adequately accomplish removing the use of JRPUMP while keeping to the calibration process. Inadequate technical rigor used when the site procedure was revised, which led to the technical errors.

Corrective Actions:

Revise the site procedure to correct the conditions identified associated with gaps and errors in the procedure. Due Date 06/30/2022. Completion tracked by Corrective Action Program

Reperform Jet Pump Calibration. Due Date 07/29/2022. Completion tracked by Corrective Action Program.

Correct errors in jet pump calibration spreadsheet and align with procedure. Due Date 07/13/2022. Completion tracked by Corrective Action Program.

Standing Order 21-005 to establishing an administrative reduction in MFLCPR limit to 0.970. Complete.

Standing Order 21-018, to provided operators the ability to calculate total core flow using reactor recirculation drive flow indications that are not impacted by the calibration of the jet pump instrumentation. Complete.

Previous Similar Events:

None.